
Gender bias and endogenous dynamics of denunciations in shaping knowledge about the heterodox

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Abstract

This paper explores how the effectiveness of medieval inquisitions, as well as contemporary and modern knowledge about heterodox communities, was shaped by inquisitors' procedural choices and by endogenous dynamics within networks of denunciation (Estévez et al. 2024; Salihović and Estévez 2026). We conceptualise inquisitorial trials as discretionary and interactive information-gathering processes in which denunciations emerge under increasing external pressure (Corsianos 2003; Emigh et al. 2020). Through formal analysis, we show that the dynamics of coercion and denunciation, together with inquisitors' gender bias in investigative work, produced informational redundancy and skewed knowledge about the composition of heterodox communities.

The data derive from the register of the trial conducted by the inquisitor Albert of Castellario against the Waldensians in the Italian town of Giaveno in early 1335 (Merlo 1977). Information on depositions and summonses was manually coded from the register. The dataset includes the timing of denunciation and summons events, as well as the identities of deponents, the individuals they accused, congregations they attended, and those summoned by the inquisitor. We additionally coded the gender of those involved, information on kinship ties, and whether accusations were extracted under torture. The dataset comprises 110 deponents (34 women and 76 men), 239 denounced individuals, and 267 individuals in total, as well as 747 unique deponent-accused dyads (829 total accusation events, including accusations against entire households and repeated dyads) and 79 summonses. An accusation network was constructed from the deponent-accused dyads. We additionally constructed a bipartite network linking individuals to congregations and subsequently derived information on co-participation in congregations. The accusation network serves as the response variable in the analysis of coercive pressure and endogenous network dynamics. In the analysis of inquisitorial procedural choices regarding gender, the timing of summonses is used as the response variable.

Aggregate patterns in denunciations and in the timing of summonses of male and female witnesses suggest that the information gathered by the inquisitor during the trial eventually became redundant, potentially contributing to the decision to conclude the proceedings (Figure 1). The summons data further suggest that the inquisitor may have prioritised male over female witnesses. To understand the mechanisms underlying these aggregate patterns, we employed dynamic network actor models (Stadtfeld and Block 2017), counterfactual simulations, and survival models (Hosmer et al. 2008). Using network models, we examined

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endogenous mechanisms shaping deponents' decisions both to denounce and in selecting targets. Counterfactual simulations of the accusation network enabled us to compare the impact of male and female witnesses on the extent to which inquisitors uncovered members of the heterodox community. We employed proportional hazards and accelerated failure time models to examine whether the gender of suspects influenced the inquisitor's prioritisation of interrogations.

The results show that increases in inquisitorial coercion led to heightened rates of accusation and to an increasing tendency to denounce socially proximate individuals. More importantly, we find that "piling on" shaped target selection in accusations, as deponents relied on public information about who had already been summoned before the inquisitor or who had absconded from the trial (Figure 2). Counterfactual simulations further suggest that the inquisitor followed leads provided by male witnesses more attentively than those provided by female witnesses. Survival models show that the gender of suspects significantly affected the timing of summonses following a first accusation, even when controlling for other incriminating information. Men were summoned sooner than women after first being accused, suggesting that the inquisitor prioritised their testimonies over those of women (Figure 3).

References

- Corsianos, Marilyn. 2003. 'Discretion in Detectives' Decision Making and "high Profile" Cases'. *Police Practice and Research*, ahead of print, September 1. World. <https://doi.org/10.1080/15614260320001>
- Emigh, Rebecca Jean, Dylan Riley, and Patricia Ahmed. 2020. 'The Sociology of Official Information Gathering: Enumeration, Influence, Reactivity, and Power of States and Societies'. In *The New Handbook of Political Sociology*, edited by Cedric de Leon, Isaac William Martin, Joya Misra, and Thomas Janoski. Cambridge University Press. <https://doi.org/10.1017/9781108147828.012>.
- Estévez, José Luis, Davor Salihović, and Stoyan V. Sgourev. 2024. 'Endogenous Dynamics of Denunciation: Evidence from an Inquisitorial Trial'. *PNAS Nexus* 3 (9): page 340. <https://doi.org/10.1093/pnasnexus/pgae340>.
- Hosmer, David W., Stanley Lemeshow, and Susanne May. 2008. *Applied Survival Analysis: Regression Modeling of Time-to-Event Data*. John Wiley & Sons.
- Merlo, Grado Giovanni. 1977. *Eretici e inquisitori nella società piemontese del Trecento*. Claudiana.
- Salihović, Davor, and José Luis Estévez. 2026. 'Gender Bias in Medieval Inquisitions and Its Place in Shaping Knowledge about the Heterodox'. *Social Science History* 50 (1): 171–201. <https://doi.org/10.1017/ssh.2025.10099>.
- Stadtfeld, Christoph, and Per Block. 2017. 'Interactions, Actors, and Time: Dynamic Network Actor Models for Relational Events'. *Sociological Science* 4 (May): 318–52. <https://doi.org/10.15195/v4.a14>.

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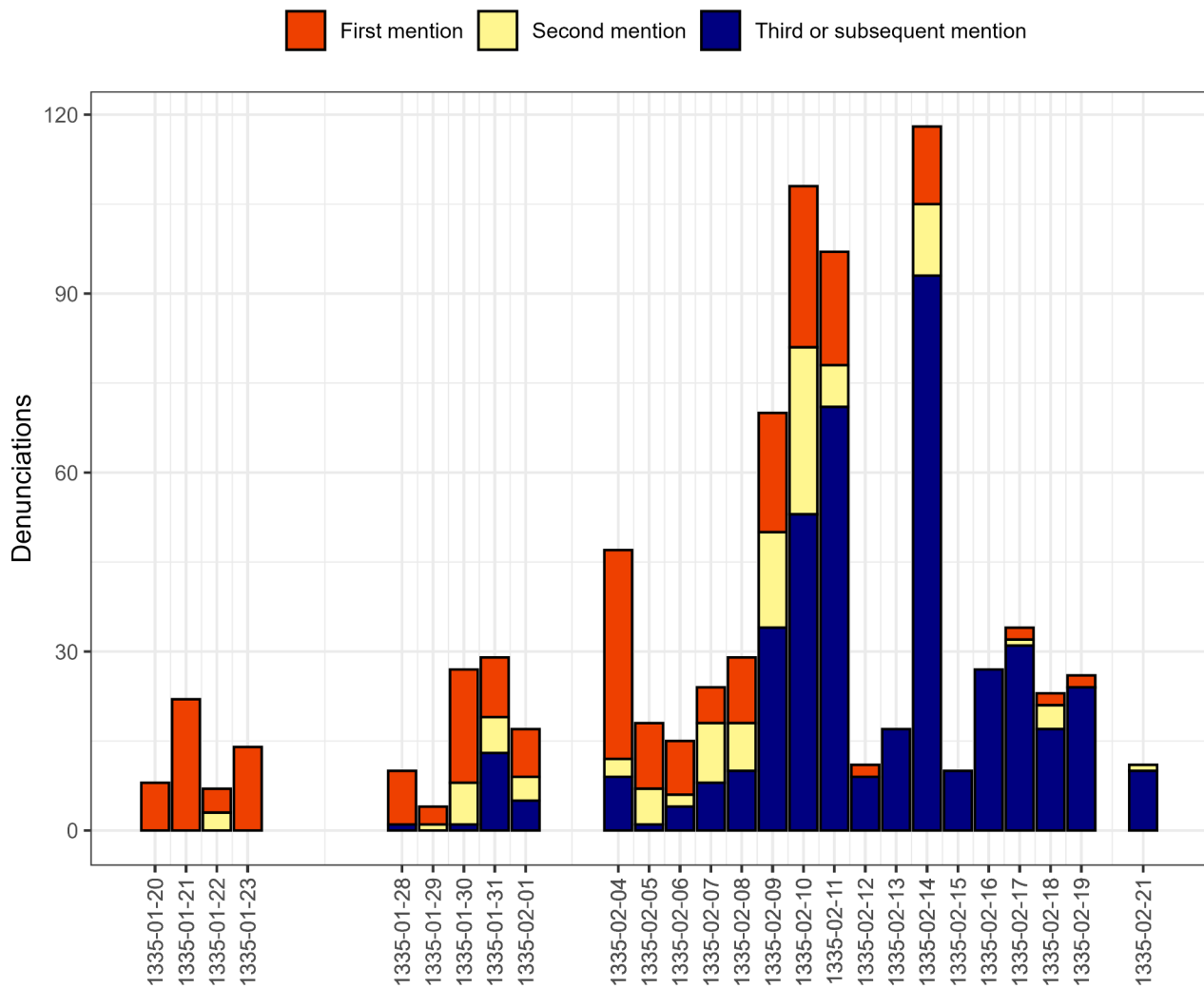


Figure 1. Redundancy in denunciations. Bars represent the number of denunciations reported to the inquisitor, categorised according to whether the accused individual was mentioned for the first time (red), second time (gold), or third and subsequent times (blue).

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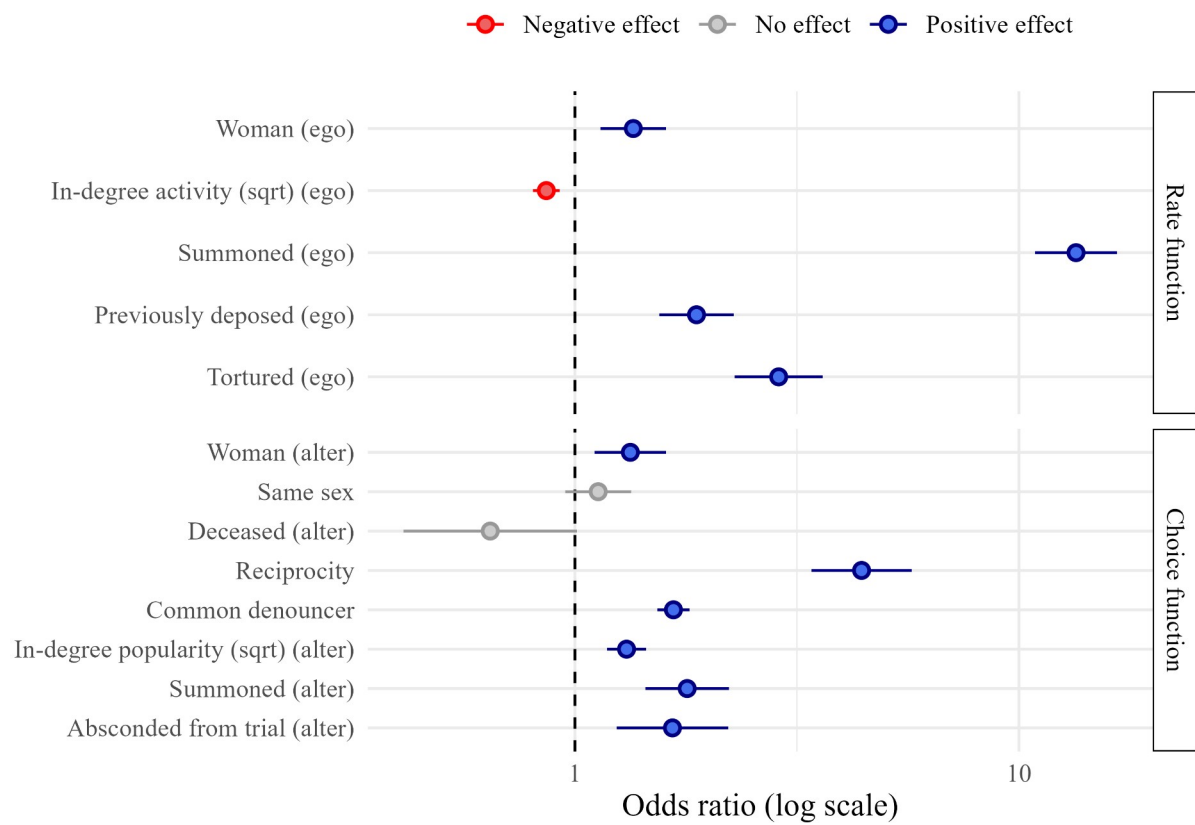


Figure 2. Odds ratios for denunciations, estimated using the dynamic network actor model. The top panel reports effects on the rate function (who engages in denunciation), while the bottom panel reports effects on the choice function (who is targeted by accusations).

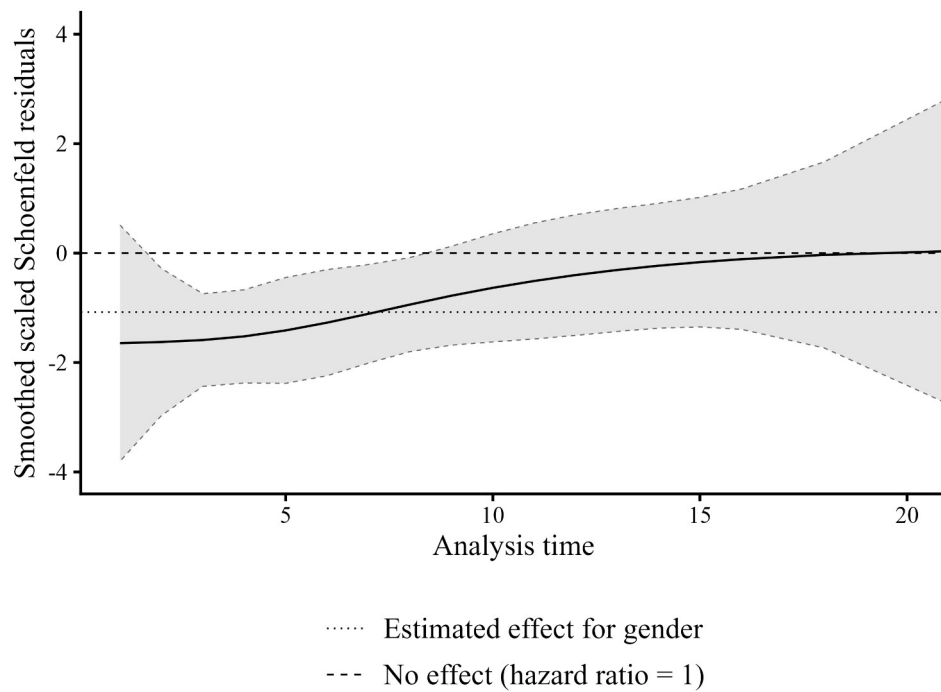


Figure 3. Smoothed Schoenfeld residuals plot from a proportional hazards model for the covariate “Gender (woman)” plotted against analysis time. The time-varying pattern suggests that the effect of gender changes over the course of the process, tending towards zero as time progresses.